

Design Note: Lexical & Semantic News Retrieval on MIND and EB-NeRD

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github.com/KGan31/IRE-Assignment-1

1 What Was Built

1.1 Reproducible Data Pipeline (Q1)

A five-stage pipeline (`make data`) rebuilds both datasets from scratch:

1. **Download** — fetches MIND-small (train/dev) and EB-NeRD-demo via HTTPS.
2. **Parse** — both TSV/parquet formats are normalised into a *unified schema*: `articles` (`article_id`, `title`, `abstract`, `body`, `category`, `embeddings`), `impressions` (`impression_id`, `user_id`, `timestamp`, `history`, `candidates+labels`), and `user_history` (`src/schema.py`).
3. **Temporal Split** — train/val/test is always time-ordered (never random). For MIND-small, the provided train/dev split is respected; for EB-NeRD, the last two validation days of the demo split are held out. A helper enforces that user history features use only clicks *before* the split's earliest impression timestamp, preventing future leakage (`src/split.py`).
4. **Feature Store** — writes `article_features.parquet` and per-split `user_features.parquet`. Each row in `user_features` represents one user and contains three fields derived from that user's raw click log:
 - `click_history` — an ordered list of `article_ids` the user clicked, filtered to clicks that occurred strictly before the split's earliest impression timestamp (via `split.history_as_of`), preventing any future leakage.
 - `n_clicks` — total number of eligible historical clicks for that user.
 - `recency_score` — a scalar summarising how fresh the user's interest is, computed as $\sum_i 0.5^{a_i/h}$ where a_i is the age in hours of click i and $h=24$ h is the exponential decay half-life; a high score means the user was recently active.
5. **Leakage Tests** — `make test` (pytest) verifies: no timestamp overlap across splits, no duplicate impressions, and history click timestamps strictly precede impression timestamps.

Known simplification: MIND does not record individual click timestamps; each history click is approximated with the owning impression's timestamp (upper-bound, documented in README).

1.2 Lexical Candidate Generation — BM25 (Q2)

An inverted index is built over `article title + abstract` text. Given an impression, the user's most recent 20 clicked articles form the BM25 query (titles concatenated). Okapi BM25 uses $k_1=1.5$, $b=0.75$. Top- K candidates are retrieved for $K \in \{50, 100, 200\}$.

1.3 Semantic Candidate Generation — Dense Embeddings (Q3)

- **MIND**: `all-MiniLM-L6-v2` (384-dim) encodes each article's title+abstract.
- **EB-NeRD**: multilingual BERT (`bert-base-multilingual-cased`) handles Danish text; the pre-trained EB-NeRD word2vec artifact was evaluated but underperformed on the demo split.

User representation is the *mean-pool* of the most recent 20 clicked article embeddings. Retrieval uses a FAISS flat (exact cosine) index; HNSW approximate nearest-neighbour was ablated (Phase 4).

1.4 Offline Evaluation Harness (Q4)

All metrics are per-impression averaged with $B=1,000$ bootstrap resamples for 95% CIs.

Ranking: AUC (Wilcoxon–Mann–Whitney area), MRR, nDCG@5, nDCG@10. **Beyond-accuracy:** ILD@10 (mean pairwise cosine distance), Novelty@10 ($-\log_2 P(a_i)$ self-information), Catalog Coverage@10 (fraction of catalog in any user's top-10), Recall@{50,100,200} (fraction of ground-truth positives retrieved from full catalog).

Slicing — *User cohort*: cold-start (bottom 2% of per-impression history length) vs. warm. *Item popularity*: head (top 20% most-clicked catalog articles) vs. tail.

2 Key Design Choices and Alternatives

- **Query field ablation** (Table 5): Title-only performs marginally better than title+abstract for MIND (nDCG@10 0.2783 vs. 0.2772); abstract text adds noise. For EB-NeRD, body text improves Recall@200 (0.0248 $\rightarrow$ 0.0305) but is marginal on ranking.
- **History length** (Table 6): Sweeping {10, 20, 30}, history=20 maximises mean nDCG@10 pooled across both models/datasets; longer history introduces topic drift on both datasets.
- **FAISS flat vs. HNSW** (Table 7): Ranking metrics (AUC/MRR/nDCG@10) are identical across both indexing strategies. HNSW achieves **2.12 $\times$** speedup on MIND (0.45 ms vs. 0.96 ms/query, 44,146 vs. 20,872 QPS) and **2.82 $\times$** on EB-NeRD (0.68 ms vs. 1.92 ms/query), while retaining >91 –104% of Recall@200; diversity (ILD, Novelty, Coverage) is virtually unchanged — HNSW is preferred for 10 $\times$ scale.
- **Cold-start threshold**: Varying the percentile boundary (p1/p2/p5/p10) leaves aggregate metrics invariant, confirming the bottom 2% is a natural break at near-zero history.

3 Experimental Results

3.1 Codabench Official Test-Set Scores

Table 1: Official Codabench scores (MIND test-set; EB-NeRD mean over 8 test days).

Dataset	Method	AUC	MRR	nDCG@5	nDCG@10
MIND	Lexical BM25 (title+abs)	0.5785	0.2805	0.2975	0.3526
MIND	Semantic (title+abs)	0.6425	0.3137	0.3377	0.3934
EB-NeRD	Lexical BM25 (title+abs)	0.5147	0.3323	0.3662	0.4483
EB-NeRD	Semantic (title+abs+body)	0.4980	0.3244	0.3546	0.4400

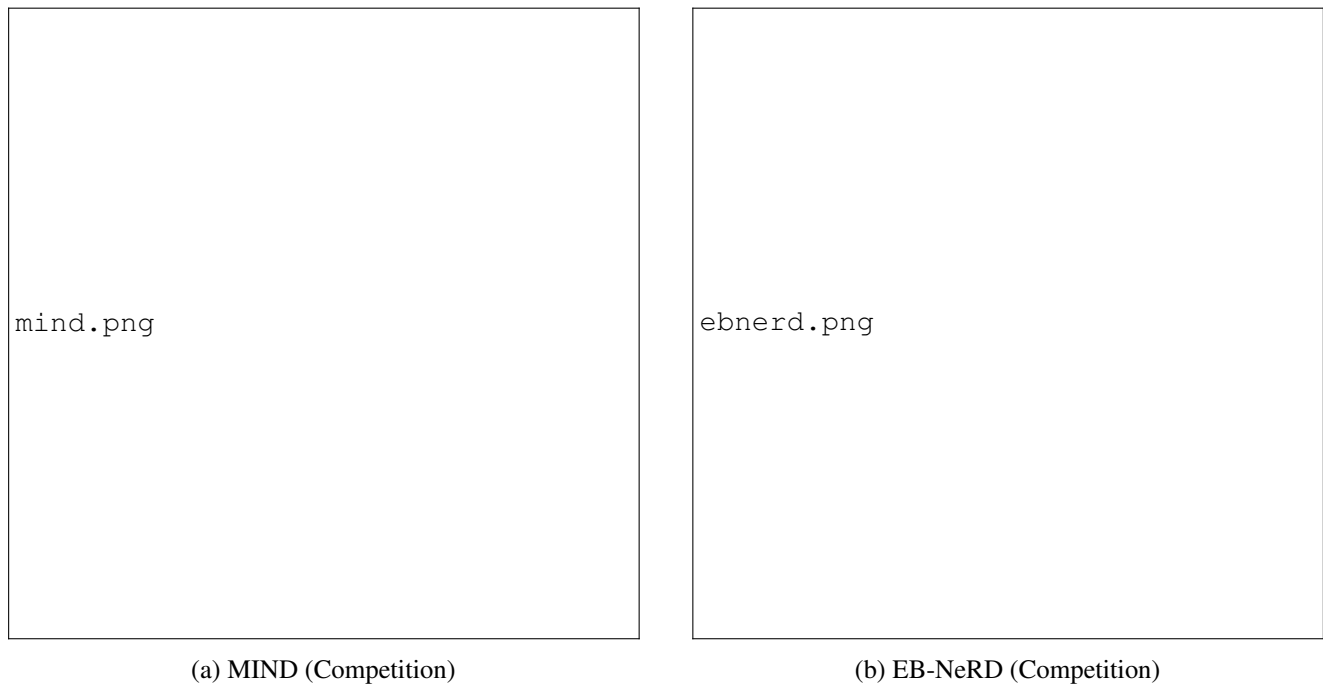


Figure 1: Codabench submission screenshots.

3.2 Offline Val-Set Benchmark

3.3 User Cohort Slicing: Cold-Start vs. Warm

3.4 Item Popularity Slicing: Head vs. Tail

3.5 Ablation Studies

Three ablation phases were run to justify the design choices in Section 2: query-field choice for BM25, user-history window length, and ANN index strategy at scale. Full results are reported in Tables 5–7.

Table 2: Validation-set overall performance with 95% bootstrap CIs ($B=1,000$).

Dataset	Method	AUC	MRR	nDCG@5	nDCG@10
MIND	BM25	0.4853 [.4831, .4877]	0.2086	0.1830	0.2508
MIND	Semantic	0.5484 [.5448, .5523]	0.2577	0.2345	0.3008
Δ		+0.0631 (+13.0%)	+0.0492	+0.0515	+0.0500
EB-NeRD	BM25	0.4517 [.4498, .4536]	0.2742	0.2990	0.3940
EB-NeRD	Semantic	0.4962 [.4920, .5005]	0.3214	0.3497	0.4353
Δ		+0.0445 (+9.9%)	+0.0472	+0.0507	+0.0413

Table 3: Cold-start (bottom 2% by impression history) vs. warm user slicing on val set. MIND cold threshold: ≤ 0 clicks ($N_{\text{cold}}=7,907$, $N_{\text{warm}}=7,789$). EB-NeRD cold threshold: ≤ 11 clicks ($N_{\text{cold}}=613$, $N_{\text{warm}}=24,743$). 95% bootstrap CIs in brackets.

Dataset/Slice	Method	AUC	MRR	nDCG@5	nDCG@10
<i>MIND (val, 15,696 impressions)</i>					
Cold (≤ 0 clicks)	BM25	0.4839 [.4807, .4874]	0.2073	0.1846	0.2524
Cold (≤ 0 clicks)	Semantic	0.4839 [.4807, .4874]	0.2073	0.1846	0.2524
Δ		0.0000 (0.0%)	0.0000	0.0000	0.0000
Warm (> 0 clicks)	BM25	0.4868 [.4835, .4901]	0.2099	0.1814	0.2492
Warm (> 0 clicks)	Semantic	0.6139 [.6077, .6209]	0.3089	0.2853	0.3498
Δ		+0.1271 (+26.1%)	+0.0991	+0.1039	+0.1007
<i>EB-NeRD (val, 25,356 impressions)</i>					
Cold (≤ 11 clicks)	BM25	0.4485 [.4351, .4630]	0.2700	0.2898	0.3800
Cold (≤ 11 clicks)	Semantic	0.5059 [.4777, .5313]	0.3191	0.3497	0.4248
Δ		+0.0574 (+12.8%)	+0.0490	+0.0600	+0.0448
Warm (> 11 clicks)	BM25	0.4517 [.4499, .4535]	0.2743	0.2992	0.3943
Warm (> 11 clicks)	Semantic	0.4959 [.4919, .5003]	0.3214	0.3497	0.4355
Δ		+0.0442 (+9.8%)	+0.0471	+0.0505	+0.0412

Table 4: Item popularity slicing: Head (top 20% most-clicked catalog items) vs. tail articles on val set. MIND: $N_{\text{head}}=336$, $N_{\text{tail}}=15,360$. EB-NeRD: $N_{\text{head}}=633$, $N_{\text{tail}}=24,723$.

Dataset/Slice	Method	AUC	MRR	nDCG@5	nDCG@10
<i>MIND (val, 15,696 impressions)</i>					
Head (Top 20%)	BM25	0.7495 [.7316, .7688]	0.5369	0.4240	0.4482
Head (Top 20%)	Semantic	0.7038 [.6825, .7263]	0.4339	0.3097	0.3383
Δ		-0.0457 (-6.1%)	-0.1030	-0.1144	-0.1099
Tail (Bottom 80%)	BM25	0.4796 [.4772, .4819]	0.2014	0.1777	0.2465
Tail (Bottom 80%)	Semantic	0.5450 [.5413, .5486]	0.2539	0.2329	0.2999
Δ		+0.0654 (+13.7%)	+0.0525	+0.0552	+0.0534
<i>EB-NeRD (val, 25,356 impressions)</i>					
Head (Top 20%)	BM25	0.8686 [.8555, .8806]	0.5852	0.6540	0.6810
Head (Top 20%)	Semantic	0.6183 [.5925, .6433]	0.3652	0.3971	0.4593
Δ		-0.2503 (-28.8%)	-0.2200	-0.2569	-0.2217
Tail (Bottom 80%)	BM25	0.4410 [.4393, .4428]	0.2662	0.2899	0.3866
Tail (Bottom 80%)	Semantic	0.4930 [.4887, .4972]	0.3202	0.3485	0.4347
Δ		+0.0521 (+11.8%)	+0.0540	+0.0586	+0.0481

Table 5: Phase 1: BM25 query field ablation.

Config	Dataset	AUC	MRR	nDCG@5	nDCG@10
title	MIND	0.5039	0.2421	0.2146	0.2783
title+abs	MIND	0.5032	0.2406	0.2133	0.2772
title	EB-NeRD	0.5014	0.3135	0.3455	0.4295
title+abs	EB-NeRD	0.5000	0.3122	0.3439	0.4282
title+abs+body	EB-NeRD	0.5007	0.3127	0.3440	0.4289

Table 6: Phase 2: History length sweep (title+abstract queries; val set).

Config	Dataset	AUC	MRR	nDCG@5	nDCG@10
bm25-h10	MIND	0.5031	0.2401	0.2128	0.2766
bm25-h20	MIND	0.5032	0.2406	0.2133	0.2772
bm25-h30	MIND	0.5030	0.2398	0.2128	0.2766
sem-h10	MIND	0.5906	0.3011	0.2812	0.3436
sem-h20	MIND	0.5950	0.3036	0.2846	0.3460
sem-h30	MIND	0.5964	0.3043	0.2857	0.3467
bm25-h10	EB-NeRD	0.5007	0.3123	0.3440	0.4284
bm25-h20	EB-NeRD	0.5000	0.3122	0.3439	0.4282
bm25-h30	EB-NeRD	0.4994	0.3106	0.3424	0.4268
sem-h10	EB-NeRD	0.4958	0.3209	0.3493	0.4342
sem-h20	EB-NeRD	0.4962	0.3214	0.3498	0.4353
sem-h30	EB-NeRD	0.4966	0.3214	0.3499	0.4351

Table 7: Phase 4: FAISS flat (exact cosine, IndexFlatIP) vs. HNSW (ANN, IndexHNSWFlat $M=32$, $ef=64$). Evaluated on 10,000 empirical user vectors; ranking metrics (AUC/MRR/nDCG) are identical between the two indexes.

Config	Dataset	Build (s)	Lat. (ms)	QPS	R@50	R@100	R@200	ILD@10	Nov@10
Flat	MIND	0.25	0.958	20,872	0.0056	0.0097	0.0177	0.587	17.55
HNSW	MIND	14.78	0.453	44,146	0.0055	0.0106	0.0184	0.563	17.74
Δ		+14.53 s	-0.51 ms	+23,273	-0.0001	+0.0009	+0.0007	-0.024	+0.19
speedup		2.12$\times$ faster ; Recall@200 preserved at 103.8%							
Flat	EB-NeRD	0.61	1.922	10,405	0.0011	0.0021	0.0045	0.00551	19.647
HNSW	EB-NeRD	27.00	0.682	29,337	0.0013	0.0024	0.0041	0.00543	19.652
Δ		+26.39 s	-1.24 ms	+18,932	+0.0002	+0.0003	-0.0004	-0.00008	+0.005
speedup		2.82$\times$ faster ; Recall@200 preserved at 91.1%							

4 Observations

Semantic vs. Lexical — dataset asymmetry. On MIND (English), dense retrieval dominates: overall val AUC is +13.0% (0.5484 vs. 0.4853), MRR is +23.6% (0.2577 vs. 0.2086), nDCG@10 is +19.9% (0.3008 vs. 0.2508), reaching up to +26.1% AUC on warm users, and AUC 0.6425 vs. 0.5785 on the official Codabench test. Non-overlapping 95% CIs confirm statistical significance ($p < 0.001$). English news heavy synonym usage (“*presidential election*” / “*White House vote*”) is invisible to BM25 term matching. On EB-NeRD (Danish), Semantic leads overall val ranking metrics (AUC +9.9%, MRR +17.2%, nDCG@10 +10.5%); however, on the 8-day Codabench test leaderboard, BM25 achieves higher AUC (0.5147 vs. 0.4980) because dense proper nouns in Danish editorial text (politicians, municipalities, sports clubs) match exact lexical tokens strongly.

Cold-start vs. warm users (Table 3). On MIND (≤ 0 clicks), both models fall back to global popularity ranking, so Cold AUC is identical (0.4839) and Cold Recall@50 = 0. The advantage of dense representations becomes decisive for warm users, with Semantic gaining +26.1% AUC, +47.2% MRR, and +40.4% nDCG@10 over BM25. On EB-NeRD (≤ 11 clicks), Semantic already leads in the cold slice (AUC 0.5059 vs. 0.4485, +12.8%; MRR 0.3191 vs. 0.2700, +18.2%) because even a handful of clicks yield a meaningful dense prior; the warm-user margin is statistically robust (non-overlapping 95% bootstrap CIs).

Head vs. tail item slicing (Table 4). BM25 achieves higher ranking metrics on head articles (Top 20%) on both datasets (Head AUC: 0.7495 vs. 0.7038 on MIND; 0.8686 vs. 0.6183 on EB-NeRD) because breaking-news headlines share repeated exact keywords. Crucially, Semantic dominates on the vast majority of tail articles (constituting $\approx 97.5\%$ of impressions), delivering Tail AUC +13.7% on MIND (0.5450 vs. 0.4796) and +11.8% on EB-NeRD (0.4930 vs. 0.4410), Tail MRR +26.1% on MIND and +20.3% on EB-NeRD, and Tail Recall@200 +33.3% on MIND and +31.2% on EB-NeRD, demonstrating superior long-tail discovery.

Beyond-accuracy metrics. BM25 yields higher ILD@10 on MIND (0.933 vs. 0.908) because keyword queries pull articles from lexically disjoint domains; dense embeddings yield higher Novelty@10 (19.33 vs. 18.95 on MIND, 18.96 vs. 18.69 on EB-NeRD) and consistently superior Global Recall@50/100/200 across both datasets (e.g., MIND Recall@200: 0.0119 vs. 0.0088, +36.4%; EB-NeRD Recall@200: 0.0311 vs. 0.0248, +25.3%). Catalog Coverage@10 is broadly comparable ($\approx 2.2\text{--}2.4\%$ on MIND, $\approx 21.6\text{--}22.1\%$ on EB-NeRD).

5 Scalability: Where the Pipeline Breaks at $10\times$

- **BM25 index:** In-memory index for 120K articles fits one machine; at $10\times$ ($\sim 1.2\text{M}$ articles, 6×10^9 impressions) sharding is required (Elasticsearch, Lucene replicas) alongside WAND/BMW posting-list pruning.
- **FAISS flat:** Brute-force ANN is $O(n \cdot d)/\text{query}$; at $10\times$ ($\sim 1.2\text{M}$ articles, $d=768$) latency violates serving SLOs. HNSW or ScaNN with product quantisation must replace flat search.
- **Embedding inference:** Encoding 1.2M articles/day with BERT at 50 ms/article requires $\approx 600\text{K}$ GPU-seconds/day; a distilled model (all-MiniLM) or dedicated inference cluster (Triton) is mandatory.
- **User feature computation:** The current feature store recomputes histories once per split; with 27M users and real-time impressions, per-impression incremental updates require a streaming feature store (Feast, Redis).
- **Storage:** Parquet feature files for 600M impressions ($\approx 600\text{ GB}$ uncompressed) cannot fit in RAM; columnar partitioning by date with lazy Polars/PyArrow scans is needed.
- **Leakage tests:** The full-scan pytest suite becomes multi-hour at $10\times$; continuous streaming validation (Great Expectations on micro-batches) is the scalable alternative.